Project Time Management

Topic # 5

Chapter 6 – PMBOK

Chapter 6 - Schwalbe

Project Plan

- A detailed plan, however, must also include a schedule indicating the start and completion times both for project as a whole and for individual activities.
- A detailed project plan will enable us to
  - Ensure availability of resources when required.
  - Produce a detailed schedule showing which staff carry out each activity.
  - Avoid different activities competing for the same resources at the same time.
  - Produce a detailed plan against which actual achievement may be measured.

Scheduling

- 'Time is nature's way of stopping everything happening at once'
- Having
  - worked out a method of doing the project
  - identified the tasks to be carried
  - assessed the time needed to do each task
- Need to allocate dates/times for the start and end of each activity

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Main Processes in Project Time Management:

- Identifying the specific activities that the project team members and stakeholders must perform to produce the project deliverables.
  - *An **activity** or **task** is an element of work normally found on WBS that has an expected duration, a cost, and resource requirements.*
- Identifying & documenting the relationships between project activities.
- Estimating how many & type of resources to complete the activity
- Estimating the number of work periods that are needed to complete individual activities.
- Creation of project schedule using activity sequencing, activation duration estimates & resource requirements.
- Involves managing & controlling project schedules.

Scheduling (Activities)

- A software project is accomplished or built through a number of smaller tasks.
- While some of these tasks can be completed at their own pace without affecting the project as a whole, there are certain tasks, which are very critical.
- These critical tasks have a deep impact on the project and if they fall behind schedule, the entire project is affected and will be delayed.
- It is the responsibility of the ***project manager*** to identify the critical tasks and later track their progress so that any delay is recognized and corrected immediately.
- The project manager can achieve this provided he has a schedule that will enable him to monitor and track the progress of the project.
- Such a schedule is termed as a ***software project schedule***.

Importance of Project Schedules

- Managers often cite delivering projects on time as one of their biggest challenges
- Time has the least amount of flexibility; it passes no matter what happens on a project
- Schedule issues are the main reason for conflicts on projects, especially during the second half of projects

Objectives of Activity Planning

These help us to:

- **Assess** the **feasibility** of the planned project completion date
- **Identify** when **resources** will need to be deployed to activities
- **Calculate** when **costs** will be incurred

This helps the **co-ordination** and **motivation** of the project team

Defining Activities

- An **activity** or **task** is an element of work normally found on the work breakdown structure (WBS) that has an expected duration, a cost, and resource requirements
- Activity definition involves developing a more detailed WBS and supporting explanations to understand all the work to be done so you can develop realistic cost and duration estimates

Projects and Activities
A - Defining Activities

- A project is composed of a number of interrelated activities.
- A project may start when at least one of its activities is ready to start.
- A project will be completed when all of its activities have been completed.
- An activity must have clearly defined start and end-points.
- Resources required by the activity must be forecastable to measure the effort.
- The duration of the activity must be forecastable.
- Some activities might require that others are completed before they can begin (*these are known as precedence requirements*)

Scheduling – Activities

- ***Generating Activity List***
- Assigning and Knowing Activity Attributes
- ***Defined outcomes***
  - Every scheduled task should have a specific outcome. Outcome could be products, also called deliverables.
- ***Milestone*** – an event signifying the completion of a major project deliverable.
- **Milestone** – an event signifying the completion of a major project deliverable. A **milestone** on a project is a significant event that normally has no duration.
  - Milestones versus Deliverable
- It often takes several activities and a lot of work to complete a milestone
- They're useful tools for setting schedule goals and monitoring progress
- Examples include obtaining customer sign-off on key documents or completion of specific products

Sequencing Activities

- Involves reviewing activities and determining dependencies
- A **dependency** or **relationship** is the sequencing of project activities or tasks
- You *must* determine dependencies in order to use critical path analysis

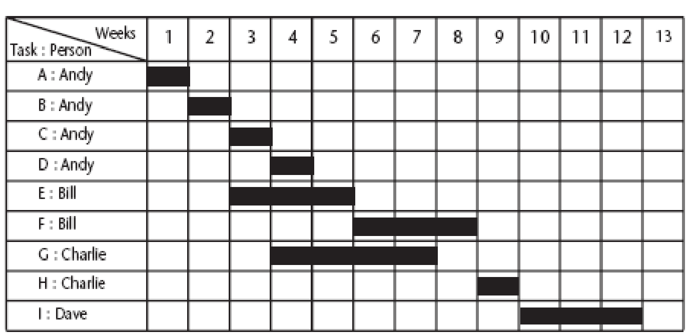
Activity Sequencing

- A dependency or relationship relates to the sequencing of project activities or tasks.
- Basic Reasons for Dependencies:
  - Mandatory Dependencies
  - Discretionary Dependencies: Defined by project teams
  - External dependencies: Involves relationship b/w project & non project activities

Sequencing and Scheduling Activities

Simple Sequencing Technique

One way of presenting a project plan is to use a bar chart

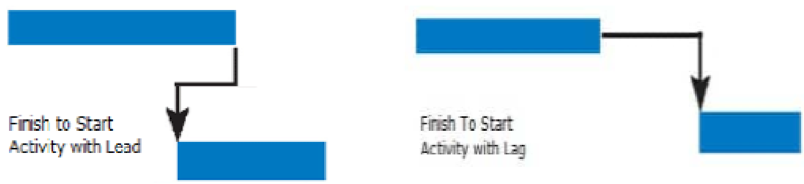


Activity key
A : Overall design
B : Specify module 1
C : Specify module 2
D : Specify module 3
E : Code module 1
F : Code module 3
G : Code module 2
H : Integration testing
I : Sytem tesing

- The chart tells us **who** is doing **what** and **when**.
- But it does not shows why certain decisions are made.
- No logical relations.

Lead & Lag Time

- ***Lead Time***: When first activity is still running the second activity starts, the remaining time left for first activity is lead time.
- ***Lag Time***: When there is a delay or wait period for the second activity to start even after the finish time of the predecessor it is called lag time.

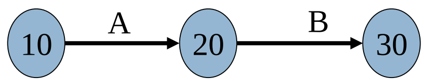


Network Diagram

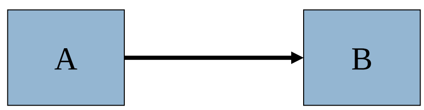
- A task network also called an ***activity network*** is a graphical representation of the task flow of the project.
- Helps to identify task sequences, interdependencies & parallel activities
- Identify critical path tasks/activities, tasks that must be completed on schedule if the project as a whole has to be completed on schedule
- Predecessor v/s Successor Activity
- Network Diagrams
  - Activity on Node (AON)
  - Activity on Arrow (AOA)

AON V/S AOA

- ***Activity on Arrow***

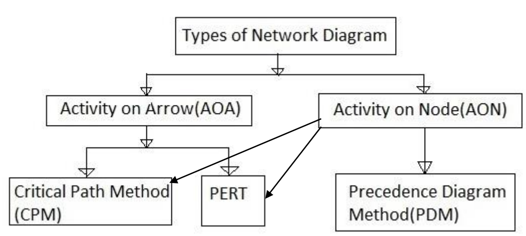


  - Arrow represents activities, nodes represents events
- ***Activity on Nodes***



  - Nodes represents activities, arrow represents order of activities.

Network Planning Models



AOA approach is used to visualize the project as a network where activities are drawn as arrows joining circles, or nodes, which represent the possible start and/or completion of an activity or set of activities.

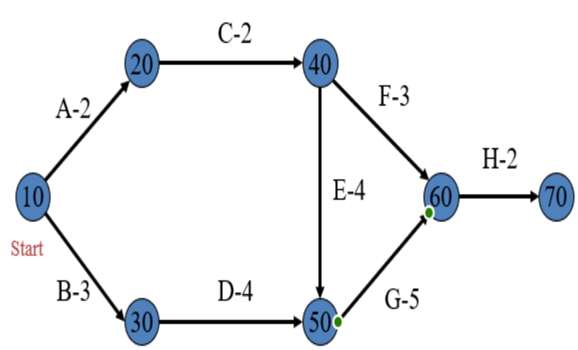
AON networks, activities are represented as nodes and links between nodes represent precedence (or sequencing) requirements. Majority of the computer applications use this method.

Critical Path & Slack

- ***Critical path*** – the sequence of dependent tasks that determines the earliest possible completion date of the project.
- ***Slack or Float time*** – the amount of delay that can be tolerated between the starting time and completion time of a task without causing a delay in the completion date of the entire project.

Example

Code	Time (weeks)	Predecessor
A	2	-
B	3	-
C	2	A
D	4	B
E	4	C
F	3	C
G	5	D,E
H	2	F,G

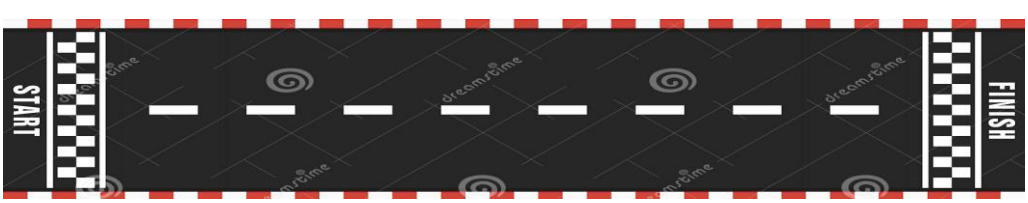


Network Models & Critical Path

- Find All Paths & Duration (ACFH, ACEGH, BDGH)
- Critical Path,
- Project Completion Time,
- Critical Activities
- Project completion 15 weeks

Start Times & Finish Times

- Earliest Start Time (EST) & Earliest Finish Time (EFT)
  - Assume all the activities start as soon as all predecessors complete
  - Early start time should be calculated considering the last activity early finish time whereas finish time can be calculated by adding the duration of the activity to the current activity start time
- Late Start Time (LST) & Late Finish Time (LFT)
- LST for an activity is minimum of LFT minus the duration, for all of its succeeding activities
- –LFT = LST of all successor activity – 1



Forward Pass

- **Step 1:** Make a forward pass through the network as follows: For each activity *i* beginning at the Start node, compute:
  - **Earliest Start Time (ES)** = the maximum of the earliest finish times of all activities immediately preceding activity . (This is 0 for an activity with no predecessors.). This is the earliest time an activity can begin without violation of immediate predecessor requirements.

    Earliest Start Time (ES) = the **earliest time** that an **activity** can **begin**
  - **Earliest Finish Time (EF)** = This represent the earliest time at which an activity can end.

    Earliest Finish Time (EF) = the **earliest time** that an **activity** can **be completed**

The project completion time is the maximum of the Earliest Finish Times at the Finish node.

Backward Pass

Step 2: Make a backwards pass through the network as follows: Move sequentially backwards from the Finish node to the Start node. At a given node, *j*, consider all activities ending at node *j*. For each of these activities, (*i*,*j*), compute:

- **Latest Finish Time (LF)** = the minimum of the latest start times beginning at node. This is the latest time an activity can end without delaying the entire project.

  he **earliest time** than an **activity** can **be completed without *lengthening* the minimum project duration**
- **Latest Start Time (LS)** = This is the latest time an activity can begin without delaying the entire project.

  the **latest time** that an **activity** can **begin without *lengthening* the minimum project duration**

Project Duration

Duration of the project is the **difference** between the ***maximum*** **value of the "latest finish time"** of projects and the ***minimum*** **value of the "earliest start time"** of projects

```
Project duration = max(LF) - min(ES)
```

Total Float & Free Float

- ***Total Slack or Float*** is the amount of time a task can be delayed without delaying the project's finish date.
  - ES- LS or EF-LF

A critical path is a path of activities, from the Start node to the Finish node, with 0 slack times.

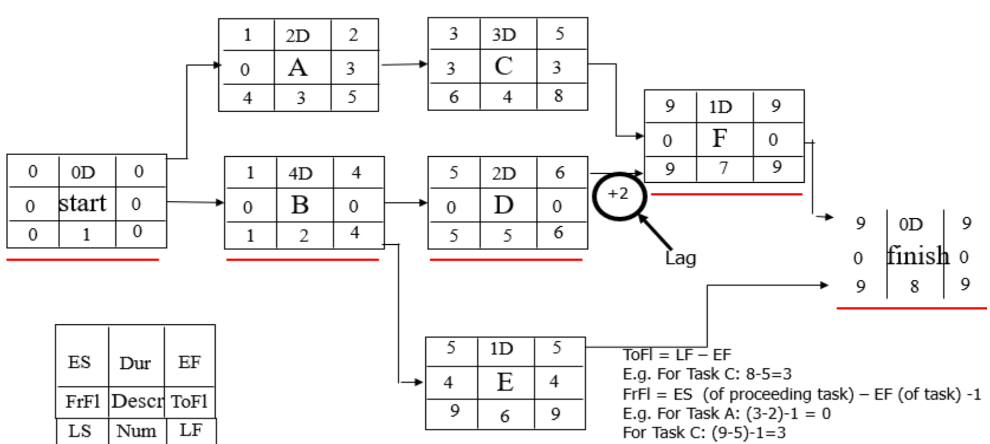
Critical activity = an **activity** in the project with **slack = 0** i.e.: **critical activities** have **no slack**, when such project is **delayed**, the **duration of the project will** ***increase***

- ***Free Slack*** is the amount of time that a task can be delayed without delaying the earliest start time of any successor task.
  - Calculated by difference between early start of next and early finish of current

Example

Activity	Estimated Duration (Days)	Constraints
A	2	Can start at any time
B	4	Can start at any time
C	3	Cannot start until A has ended
D	2	Cannot start until B has ended
E	1	Cannot start until B has ended
F	1	Cannot start until C has ended and 2 days after D has ended

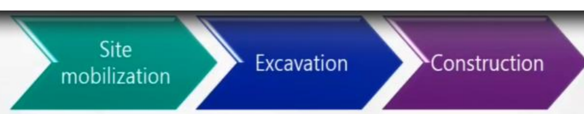
Solution Using Activity on Node



PDM (Precedence Diagramming Method) OR AON

- Uses boxes or rectangles referred as nodes to represent activities.
- Arrows show relationships between activities.
- This technique is also called Activity-On-Node (AON)
- PDM includes four types of dependencies or logical relationship.

CRITICAL PATH METHOD



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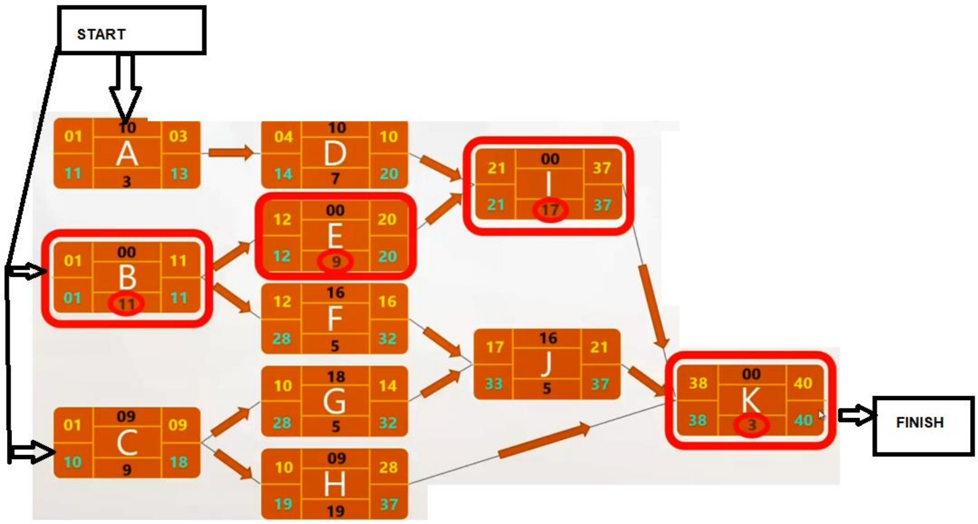
ACTIVITIES	DURATION	PREDECESSOR
A	3	-
B	11	-
C	9	-
D	7	A
E	9	B
F	5	B
G	5	C
H	19	C
I	17	D,E
J	5	F,G
K	3	I,J,K

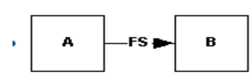
ES	Early start	It is the earliest time that an activity can start
EF	Early finish	It is the earliest time that an activity can finish
LS	Late start	It is the late time that an activity can start
LF	Late finish	It is the late time that an activity can finish
D	Duration	Duration of activities
F	Float	The float for an activity is the amount that its duration can slip without causing the project to be delayed

Solution

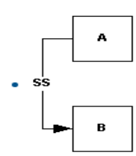


Activity Relationships

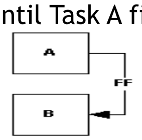
- **Finish to Start:** Activity A must be completed before Activity B starts



- **Start to Start:** Task A cannot Start until Task B starts

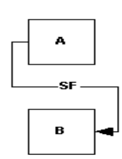


- **Finish to Finish:** Task B cannot finish until Task A finishes

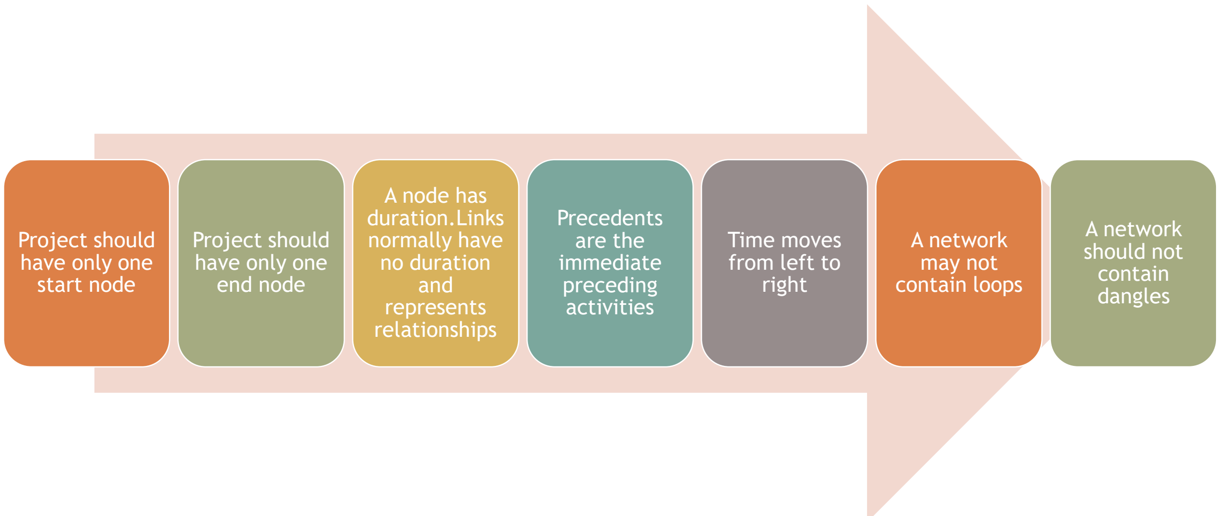


- **Start to Finish:** Task B cannot finish until Task A starts

(Rarely used)



Recap: Constructing Network Diagrams



END OF TOPIC 5A

- COMING UP!!!!!!
- Time & MS Project
- Project Lifecycles
- HR - Project Team Organization
- Project Cost Management
- Project Control
- Mid II

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